

Tharun V. Puthanveettil

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Human-Inspired Robot Learning | Multimodal Learning | Human-Robot Interaction

Research Focus

My research takes inspiration from how people learn, adapt, and develop skills to build robots that improve through experience. My long-term aim is to develop algorithms that enable robots to understand human intent with the contextual sensitivity people use to understand one another, and to act or learn from that understanding. I pursue this through policy understanding, continual adaptation, embodied experience, predictive models, multimodal human-robot interaction, and evaluation on physical systems.

[\[RESEARCH PORTFOLIO\]](#)

Research interests: human-inspired robot learning; continual and interactive learning; policy understanding and representation learning; vision-language-action policies; world and inverse-dynamics models; multimodal HRI; compliant manipulation; real-to-sim-to-real transfer; robot-policy evaluation.

Current Research Program

Policy Understanding and Representation Analysis

EXPERIMENTS IN PROGRESS

What has a policy learned, and which observations, prompts, states, and priors shape its actions?

- Uses attention maps, Grad-CAM, state-vector ablations, prompt comparisons, and pretraining/self-supervised representation comparisons as behavioral diagnostics rather than causal explanations. [\[PUBLIC OVERVIEW\]](#)

Continual and Human-Guided Adaptation

SYSTEM IMPLEMENTED; EVALUATION MANUAL

How can robots improve from intervention and correction without discarding prior capability?

- Connects takeover, correction capture, data review, conversion, retraining, testing, and controlled rollout through common data and policy interfaces; automated policy evaluation is not yet complete. [\[PUBLIC OVERVIEW\]](#)

Predictive Models, Inverse Dynamics, and Transfer

OFFLINE EVALUATION COMPLETED

How can predicted outcomes and inverse dynamics support action learning on physical robots?

- Adapted a published video-action architecture by retraining its action head on teleoperation data and completed offline evaluation. Physical validation and simulator-trained inverse dynamics remain future work. [\[PUBLIC OVERVIEW\]](#)

Egocentric Experience and Multimodal Interaction

INTERFACE IN DEVELOPMENT

How can embodied human experience be converted into useful robot learning signals?

- Developing a portable egocentric demonstration interface that reuses the synchronization, review, and dataset-conversion infrastructure of the existing robot-learning platform. [\[PUBLIC OVERVIEW\]](#)

Education

Master of Engineering in Robotics, GPA: 3.9/4.0

University of Maryland, College Park

Aug. 2022 - May 2024

Maryland, USA

B.Tech. in Electronics and Communication Engineering, GPA: 8.59/10.0

Vellore Institute of Technology

Jul. 2014 - Apr. 2018

India

Technical Skills

Robot learning: PyTorch, TensorFlow, LeRobot, action-chunking policies, vision-language-action models, diffusion/flow policies, imitation learning, self-supervised learning

Robotics and control: ROS 2, ROS, MoveIt 2, impedance/compliance control, visual servoing, behavior trees, bimanual manipulation, force-torque feedback

Simulation and perception: Isaac Sim/Lab, MuJoCo, Gazebo, OpenCV, Open3D, RGB-D cameras, detection, segmentation, tracking, pose estimation

Programming and systems: Python, C++, SQL, Linux, Docker, Git, REST APIs

Intellectual Property, Mentoring, and Communication

Patents: Inventor on three filed patent applications; details withheld pending publication.

Research supervision: Supervised an engineering intern extending the robot-learning platform for compliant bimanual data collection, learning, analysis, and rollout.

FIRST Robotics: Electrical Programming Head Coach, The Irrational Engineers (Jun. 2024–Present); mentor high-school students in controls, computer vision, and software. Write about robot learning, embodied intelligence, and credible research practice.

[\[HUMAN IN THE LOOM\]](#) [\[ESSAY ARCHIVE\]](#)

Research and Professional Experience

Robotics Engineer - Robot Learning and Deformable Manipulation

Jun. 2024 - Present

General Motors R&D

Michigan, USA

- Architected and implemented a modular ROS 2 robot-learning platform connecting bimanual teleoperation, multimodal perception, synchronized data capture, simulation, dataset conversion, policy training, manual evaluation, staged testing, and physical deployment. [\[PUBLIC OVERVIEW\]](#)
- Developed the continual-learning workflow from human takeover and correction capture through review, retraining, testing, and controlled rollout; designed common data, policy, and deployment interfaces so each stage can be studied independently.
- Studied input and representation influence in trained policies using attention, Grad-CAM, state-vector ablations, prompt comparisons, and pretrained/self-supervised representation comparisons.
- Built task-conditioned detection, segmentation, tracking, and pose-servoing services with foundation models; reduced perception latency by approximately 35% and sustained 30–40 FPS through embedding reuse and temporal mask propagation.
- Developed force-aware bimanual manipulation of deformable linear objects using impedance control and force-torque feedback; quantified a twofold increase in failures without elastic-tension regulation.
- Created a digital twin and behavior-tree-driven synthetic-data workflow to characterize policy degradation under sensing, timing, dynamics, actuation, contact, and visual shift.
- Architected the Cartesian motion-policy, safety, and evidence pipeline; supervised its extension to dual-arm collection and rollout with admittance-based compliance, synchronized RGB-wrench data contracts, matched collection/deployment control, and measured-state reseeding.
- Evaluated action-chunking and open-source vision-language-action policies while developing measures of partial progress, failure type, recovery, repeatability, and compliance sensitivity.

Researcher - AI and Robotics

Feb. 2023 - Mar. 2024

CATT Laboratory, University of Maryland

Maryland, USA

- Served as core architect of a multimodal sequential anomaly-detection pipeline using modality-specific LiDAR, odometry, and network encoders with reconstruction-based scoring. The published system reported 98% accuracy versus 72% for LiDAR alone, and 97% with half the training data. [\[PUBLICATION\]](#)
- Co-developed Track Anything Rapter, a ROS 2 aerial system accepting click, image, text, and box target specifications through foundation-model detection/segmentation, visual servoing, and PX4 control. [\[PAPER\]](#) [\[CODE + DEMOS\]](#)
- Served as core architect of pose-based sequential models combining human-pose structure, temporal context, and multiple views through GCN-transformer architectures; evaluated cross-view transfer, structured occlusion, keypoint removal, and temporal subsampling. Under the reported occlusion protocol, multi-view F1 reached 0.82 versus 0.76 for single-view recognition. [\[POSTER\]](#) [\[POSEFUSION\]](#) [\[FALL DETECTION\]](#)

Researcher - Collaborative Robotics and Computer Vision

Jun. 2021 - Jan. 2023

Indian Institute of Science and University of Maryland

India and USA

- Developed a vision-guided precision-weeding system combining crop/weed detection, root and plant-spread estimation, workspace adaptation, hybrid inverse kinematics, closed-loop control, and end-effector design. [\[PAPER + FIGURES\]](#) [\[DEMO\]](#) [\[CODE\]](#)
- Developed synthetic-data and augmentation pipelines for weed localization; improved robustness by 20–30% through hard-negative sampling and structured distractors. Co-designed a dual-purpose end effector and adaptive position/torque-control methods for delicate plant geometry.
- Implemented an exploratory policy combining demonstration, speech, and gesture for robot instruction. [\[NATSGD RESEARCH\]](#)
- Developed and evaluated a U-Net monocular depth model in an IROS 2021 challenge setting. [\[CODE\]](#)

Project Engineer - AI Innovation

Jul. 2018 - May 2021

CTO Office, Wipro Digital

Bangalore, India

- Served as AI project lead for applied-research prototypes in computer vision, HCI, recommendation, generative modeling, and mobile inference; coordinated quarterly prototype selection and delivery.
- Built tool tracking, mobile vehicle recognition, photogrammetry, human-pose estimation, virtual try-on, and recommendation components for industrial and retail workflows. [\[MY STYLE\]](#) [\[VIRTUAL TRY-ON\]](#) [\[DEMO\]](#)

Peer-Reviewed Publications

1. A. Zoukarni, M. Noorani, J. S. Baras, J. Mirenzi, **T. V. Puthanveettil**, and C. D. Grody. "Efficient Integration of Domain Knowledge and Data-Driven Methods for Securing Autonomous Cyber-Physical Systems." 2026. [\[RECORD\]](#)
2. **V. P. Tharun** and A. R. Chowdhury. "Vision-Based Hybrid IK Task Planning with Feedforward Neural Network for Collaborative Plant-Robot Interaction." *International Conference on Social Robotics*, 2025. [\[PAPER\]](#)
3. M. Noorani, **T. V. Puthanveettil**, A. Zoukarni, J. Mirenzi, C. D. Grody, and J. S. Baras. "Multimodal Anomaly Detection for Autonomous Cyber-Physical Systems Empowering Real-World Evaluation." *GameSec*, 2024. [\[DOI\]](#)
4. **V. P. Tharun**, P. Ramya, and S. Renuga Devi. "A Univariate Data Analysis Approach for Rainfall Forecasting." *ICCIS*, 2020. [\[RECORD\]](#)
5. **V. P. Tharun**, R. Prakash, and S. R. Devi. "Prediction of Rainfall Using Data Mining Techniques." *ICICCT*, 2018. [\[RECORD\]](#)
6. L. P. Pratap, P. M. Shailendrasingh, A. Anand, and **V. P. Tharun**. "Wall-Climbing Robot Using Soft Robotics." *ICPCSI*, 2017. [\[PUBLICATION\]](#)

Preprints, Manuscripts, Posters, and Research Outputs

1. **T. V. Puthanveettil**, A. Singh, Y. Jain, V. Bukka, and S. Arjun S. "Auto-Platoon: Freight by Example." *arXiv:2405.11659*, 2024. [\[PREPRINT\]](#) [\[CODE\]](#)
2. **T. V. Puthanveettil** and F. Obaid ur Rahman. "Track Anything Rapter (TAR)." *arXiv:2405.11655*, 2024. [\[PREPRINT\]](#) [\[CODE + DEMOS\]](#)
3. "Pose Fusion - Multi-View Pose Integration for Comprehensive Action Recognition." Project poster and preprint. [\[POSTER\]](#) [\[PREPRINT\]](#)
4. **V. P. Tharun** et al. "Application of Mobile Collaborative Robot using Deep Learning in Precision Weed Control." Manuscript, 2021. [\[MANUSCRIPT\]](#)
5. **V. P. Tharun** et al. "Event-Based Dynamic Obstacle Avoidance in Outdoor Environments." IROS, 2021, poster. [\[POSTER\]](#)
6. R. Prakash, **V. P. Tharun**, and S. R. Devi. "A Comparative Study of Various Classification Techniques to Determine Water Quality." 2018. [\[RECORD\]](#)
7. P. Ramya, **V. P. Tharun**, and S. Renuga Devi. "Water Quality Assessment Using Data Mining Techniques." Vellore Institute of Technology, 2018. [\[RECORD\]](#)
8. **V. P. Tharun**, P. Ramya, and S. Renuga Devi. "Rainfall Forecasting Using Time Series Analysis." 2018. [\[RECORD\]](#)

Selected Earlier Technical Projects

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|--|-------------|
| Monocular Depth Estimation <i>PyTorch, U-Net, transfer learning</i> | 2021 |
| ▪ Built a monocular depth-estimation pipeline and evaluated it in an IROS challenge setting. [CODE] | |
| Implicit Neural Representation and SLIC Segmentation <i>PyTorch, TensorFlow, OpenCV</i> | 2023 |
| ▪ Implemented coordinate-conditioned image outpainting and deep-feature superpixel segmentation, including controlled comparisons with simpler baselines. [INR] [SLIC] | |
| Maze Runner and Autonomous Retrieval Robot <i>ROS 2, Nav2, visual servoing</i> | 2023 |
| ▪ Built visual-cue-based maze navigation in Gazebo and an onboard-perception retrieval robot using sensor fusion and visual servoing. [NAVIGATION] [PLANNER] [ROBOT DEMO] | |
| Human-Centered Vision and Interactive AI <i>pose, WebVR, recommendation, generative vision</i> | 2019 - 2021 |
| ▪ Developed an AI cricket coach, virtual try-on system, and personalized style assistant combining pose estimation, action recognition, immersive feedback, recommendation, and generative modeling. [CRICKET COACH] [TRY-ON] [MY STYLE] | |
| Reinforcement-Learning Cricket Simulation <i>adaptive agents and simulation</i> | 2020 - 2024 |
| ▪ Built a simulated bowler agent that adapts to batting styles and field configurations. [CODE] | |